

Anna Y. Q. Ho

(Last updated: August 2026)

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PRIMARY RESEARCH INTERESTS

Stellar death (supernovae, gamma-ray bursts), transients, time-domain astronomy, high-energy astrophysics, radio and sub-millimeter observations, large surveys

EDUCATION & APPOINTMENTS

2026–Present	Richards Family Assistant Professor, Astronomy Department, Cornell
2022–Present	Assistant Professor, Astronomy Department, Cornell
2020–2022	Miller Postdoctoral Fellow, Astronomy Department, U.C. Berkeley
2020–2022	Affiliate, Lawrence Berkeley National Laboratory
2020	Ph.D., California Institute of Technology, Astrophysics • Thesis Advisor: Shri Kulkarni
2017	M.S., California Institute of Technology, Astrophysics
2014–2015	Fulbright Scholar, Max Planck Institute for Astronomy, Heidelberg, Germany • Host: Hans-Walter Rix
2014	B.S., Massachusetts Institute of Technology, Physics

AWARDS & HONORS

2026	<i>Scientific American</i> Young American Scientist Award
2026	Cottrell Scholar Award, Research Corporation for Science Advancement
2026	Richards Family Assistant Professorship, Cornell University
2026	Frontiers of Science Fellow, National Academy of Sciences
2026	Newton Lacy Pierce Prize in Astronomy, American Astronomical Society <i>Awarded annually for outstanding achievement, over the past five years, in observational astronomical research based on measurements of radiation from an astronomical object</i>
2024	Packard Fellowship for Science and Engineering <i>Awarded annually to 20 early-career scientists and engineers in the U.S.</i>
2024	Scialog Fellow, Research Corp. for Science Advancement (Early Science with the LSST)
2024	Alfred P. Sloan Research Fellow in Physics <i>Annual award for early career researchers in recognition of distinguished performance and a unique potential to make substantial contributions to their field.</i>
2021	Springer Thesis Prize, <i>In recognition of outstanding Ph.D. research in the physical sciences.</i>
2020	AAS Rodger Doxsey Travel Prize
2014–9	National Science Foundation Graduate Research Fellowship
2019	Keck Institute for Space Studies Affiliate
2017	TA Award, Caltech, <i>For being one of the highest rated TAs for the Spring 2017 term.</i>
2017	France Cordova Graduate Fellowship—Gordon Garmire Scholarship, Caltech
2014	MIT Karl Taylor Compton Prize <i>The highest awards presented by the Institute to students...in recognition of excellent achievements in citizenship and devotion to the welfare of MIT.</i>
2014	MIT Ida M. Green Fellowship (declined), <i>For the MIT Graduate Program in Science Writing</i>
2013	First Place, MIT DeWitt Wallace Prize for Science Writing for the Public
2012	MIT Burchard Scholar

SELECTED PUBLICITY

- 2026 **NRAO**, [Radio telescopes uncover “invisible” gas around record-shattering cosmic explosion](#)
- 2025 **BBC**, [The bizarre space explosions scientists can’t explain](#)
- 2025 Wrote the **Griffith Observer** article, [Chasing Cosmic Flashes](#)
- 2024 **MIT News**, [Alum first spots cosmic explosion](#)
- 2024 **Cornell Chronicle**, [Cornell astronomers on newly approved UVEX NASA mission](#)
- 2024 **National Geographic**, [What is causing these massive, mysterious explosions in space?](#)
- 2023 **Nature News**, [Mysterious Tasmanian devil space explosion baffles astronomers](#)
- 2023 **CNN**, [Stellar corpse called ‘Tasmanian devil’ reveals phenomenon astronomers have never seen](#)
- 2023 **Cornell Chronicle**, [NASA selects Cornell astronomer for ULTRASAT observatory](#)
- 2023 **The New York Times**, [A Cow, a Camel and a Finch Exploded in Space. What Is Going On?](#)
- 2021 **Quanta**, [New Kind of Space Explosion Reveals the Birth of a Black Hole](#)
- 2020 Wrote the **Scientific American** cover article, [Extreme Supernovae](#)
- 2020 **Science News**, [A weird cosmic flare called the Cow now has company](#)
- 2020 **Science Daily**, [Astronomers discover new class of cosmic explosions](#)
- 2020 **Sky & Telescope**, [Two New Beasts for an Explosive Zoo](#)
- 2019 Wrote article for the **Submillimeter Array Newsletter**, [SMA Observations of AT2018cow: A Prototype for Millimeter Time-domain Astronomy](#)
- 2019 **Science News**, [The cosmic Cow may be a strange supernova](#)
- 2019 **The Washington Post**, [Scientists had never seen anything like this supernova](#)
- 2019 **WIRED**, [We may have finally spotted a star turning into a black hole](#)
- 2018 **Nature News**, [Holy Cow! Astronomers agog at mysterious new supernova](#)

PUBLICATIONS

Count: 14 first author; 102 total [[full list on ADS](#)]; 11 white paper or science book contributions

h-index: 14 first author; 44 total

Citations: 973 first author; 13,000+ total

First author

- [1] (Invited Review) **AYQH** & Lu 2026, *Cows, Tasmanian Devils, and Other Mysteries: Diversity and Origins of Luminous Fast Blue Optical Transients*, submitted to *Reports on Progress in Physics*
- [2] **AYQH** et al. 2025, *A Luminous Red Optical Flare and Hard X-ray Emission in the Tidal Disruption Event AT2024kmq*, *ApJ*, **989**, 54
- [3] **AYQH** et al. 2023, *Minutes-duration Optical Flares with Supernova Luminosities*, *Nature*, **623**, 927
- [4] **AYQH** et al. 2023, *A Search for Extragalactic Fast Blue Optical Transients in ZTF and the Rate of AT2018cow-like Transients*, *ApJ*, **949**, 120
- [5] **AYQH** et al. 2022, *Cosmological Fast Optical Transients with the Zwicky Transient Facility: A Search for Dirty Fireballs*, *ApJ*, **938**, 85
- [6] **AYQH** et al. 2022, *Luminous Millimeter, Radio, and X-ray Emission from ZTF20acigmel (AT2020xnd)*, *ApJ*, **932**, 116
- [7] **AYQH** et al. 2020, *ZTF20aajnksq (AT2020blt): A Fast Optical Transient at $z \approx 2.9$ With No Detected Gamma-Ray Burst Counterpart*, *ApJ*, **905**, 98
- [8] **AYQH** et al. 2020, *SN2020bvc: a Broad-lined Type Ic Supernova with a Double-peaked Optical Light Curve and a Luminous X-ray and Radio Counterpart*, *ApJ*, **902**, 86

- [9] **AYQH** et al. 2020, *The Koala: A Fast Blue Optical Transient with Luminous Radio Emission from a Starburst Dwarf Galaxy at $z = 0.27$* , ApJ, **895**, 1
- [10] **AYQH** et al. 2020, *The Broad-lined Ic Supernova ZTF18aaqjovh (SN 2018bvw): An Optically-discovered Engine-driven Supernova Candidate with Luminous Radio Emission*, ApJ, **893**, 132
- [11] **AYQH** et al. 2019, *Evidence for Late-stage Eruptive Mass-loss in the Progenitor to SN2018gep, a Broad-lined Ic Supernova: Pre-explosion Emission and a Rapidly Rising Luminous Transient*, ApJ, **887**, 169H
- [12] **AYQH** et al. 2019, *AT2018cow: a luminous millimeter transient*, ApJ, **871**, 73
- [13] **AYQH** et al. 2018, *iPTF Archival Search for Fast Optical Transients*, ApJL, **854**, 13
- [14] **AYQH** et al. 2017, *Masses and Ages for 230,000 LAMOST Giants, via Their Carbon and Nitrogen Abundances*, ApJ, **841**, 40
- [15] **AYQH** et al. 2017, *Label Transfer from APOGEE to LAMOST: Precise Stellar Parameters for 450,000 LAMOST Giants*, ApJ, **836**, 5

Second or third author

[†] Projects for which I played a significant mentoring or advising role

- [15] [†] Hinds, K.-R., **AYQH** et al. 2026, *Early Multiwavelength Observations of AT 2026fgk: The Luminous Afterglow to Sub-luminous GRB 260310A, Identified Independently of a Gamma-ray Trigger*, submitted to ApJ
- [16] [†] Jayaraman, R., **AYQH** et al. 2026, *Constraints on Late-Time Flaring from Luminous Fast Blue Optical Transients using the Transiting Exoplanet Survey Satellite and the Zwicky Transient Facility*, accepted for publication in ApJ
- [17] [†] Sevilla, C., **AYQH** et al. 2026, *Multiwavelength Analysis of Six Luminous, Fast Blue Optical Transients*, ApJ, **1007**, 38
- [18] Perley, D. A., **AYQH** et al. 2026, *AT2024wpp: An Extremely Luminous Fast Ultraviolet Transient Powered by Accretion onto a Black Hole*, MNRAS, **549**, 1
- [19] Keating, G., Laskar, T., **AYQH** et al. 2026, *Rapid-response 1.3 mm Observations of GRB 260127A with the Submillimeter Array*, ApJL, **1005**, 31
- [20] [†] Schroeder, G., **AYQH** et al. 2025, *A Late-time Radio Search for Highly Off-axis Jets from PTF Broad-lined Ic Supernovae in GRB-like Host Galaxy Environments*, ApJ, **995**, 61
- [21] Fryer, C. L., Burns, E., **AYQH** et al. 2025, *Explaining Non-Merger Gamma-Ray Bursts and Broad-Lined Supernovae with Close Binary Progenitors with Black Hole Central Engine*, ApJ, **986**, 185F
- [22] [†] Li, M. L., **AYQH** et al. 2025, *The Nature of Optical Afterglows Without Gamma-ray Bursts: Identification of AT2023lcr and Multiwavelength Modeling*, ApJ, **985**, 124
- [23] [†] Srinivasaragavan, G. P., Perley, D. A., **AYQH** et al. 2025, *Multi-Wavelength Analysis of AT2023sva: a Luminous Orphan Afterglow With Evidence for a Structured Jet*, MNRAS, **538**, 351
- [24] Perley, D. A., **AYQH** et al. 2025, *The luminous, slow-rising orphan afterglow AT2019pim as a candidate moderately relativistic outflow*, MNRAS, **537**, 2362
- [25] Corsi, A., **AYQH** et al. 2023, *A search for relativistic ejecta in a sample of ZTF broad-lined Type Ic supernovae*, ApJ, **953**, 179

- [26] [†] Yao, Y., **AYQH**, et al. 2022, *The X-ray and Radio Loud Fast Blue Optical Transient AT2020mrf: Implications for an Emerging Class of Engine-Driven Massive Star Explosions*, ApJ, **934**, 104
- [27] [†] Yurk, N., Ravi, V., & **AYQH** 2022, *Models of Millimeter and Radio Emission from Interacting Supernovae*, ApJ, **934**, 5
- [28] Margalit, B., Quataert, E., & **AYQH** 2022, *Optical to X-Ray Signatures of Dense Circumstellar Interaction in Core-collapse Supernovae*, ApJ, **928**, 122
- [29] Perley, D. A., **AYQH** et al. 2021, *Real-time discovery of AT2020xnd: a fast, luminous ultraviolet transient with minimal radioactive ejecta*, MNRAS, **508**, 5138
- [30] Duffell, P. C. & **AYQH** 2020, *How Dense a CSM is Sufficient to Choke a Jet?*, ApJ, **900**, 193
- [31] Szkody, P., Diczynski, B., **AYQH**, et al. 2020, *Cataclysmic Variables from the First Year of the Zwicky Transient Facility*, AJ, **159**, 198
- [32] Casey, A.R., **AYQH**, et al. 2019, *Tidal interactions between binary stars drives lithium production in low-mass red giants*, ApJ, **880**, 125

White Paper & Science Book Contributions

- [1] ALMA2040 White Paper on Cosmic Explosions (2025)
- [2] **AYQH** et al. 2025, *Probing GRB reverse shocks in the far-IR*, PRIMA GO Science Book Vol. 2
- [3] **AYQH** et al. 2025, *PRIMA Follow-up Observations of Transients Discovered by Mm/submm Surveys*, PRIMA GO Science Book Vol. 2
- [4] Burns, E. et al. 2025, *Multidisciplinary Science in the Multimessenger Era* (FBOT Section)
- [5] Burns, E. et al. 2025, *Multidisciplinary Science in the Multimessenger Era* ([arXiv2502.03577](https://arxiv.org/abs/2502.03577)) (FBOT Section)
- [6] Hui, M., et al. 2024, *Moon Burst Energetics All-sky Monitor: A Beyond Earth-orbit Gamma-ray Burst Observatory*, 38th International Cosmic Ray Conference, 877
- [7] [Gamma-ray Transient Network Science Analysis Group Report](#) (Dirty Fireballs, FBOTs)
- [8] CfA Decadal White Paper, *Time-domain Astrophysics with the Submillimeter Array*
- [9] Abazajian, K., et al. 2022, *Snowmass 2021 CMB-S4 White Paper*, arXiv:2203.08024
- [10] DSA-2000 [Community Science Book](#) (Relativistic Stellar Explosions)
- [11] [PRIMA General Observer Science Book](#) (Relativistic Stellar Explosions)

FUNDING

Total: \$2.5 million

2027–2028	JWST Cycle 5 (\$162,624)
2026–2029	NSF Astronomy and Astrophysics Research Grants (\$273,846)
2026–2029	Cottrell Scholar Award (\$120,000)
2026–2027	HST Cycle 33 (\$43,676)
2026–2031	Private Gift (\$600,000)
2026	ATNF Visiting Scientist (\$2,000)
2025–2026	Research Corporation for Science Advancement Scialog Award (\$60,000)
2024–2029	Packard Fellowship for Science and Engineering (\$875,000)
2024	PCCW Frank H.T. Rhodes Leadership Grant and Mission Grant (\$5,000)
2024–2026	HST Cycles 31 & 32 (\$37,593)
2024–2026	Sloan Fellowship (\$75,000)

2023–2026	NASA Ultraviolet Transient Astronomy Satellite Participating Scientists Program (PI; \$210,000)
2023–2024	NASA <i>Swift</i> Cycle 19 (PI; \$36,000)
2023	Cornell Roger and Mary Lou West Undergraduate Research Fellowship (\$6,500)

SUCCESSFUL OBSERVING PROPOSALS

Summary: 99 successful proposals as PI (or with student/postdoc under my direct supervision as PI) for 37 distinct programs. 714.5 hr on millimeter telescopes, 714.6 hr on radio telescopes, 118.4 hr on UVOIR telescopes, 365 ksec (100 hr) on X-ray telescopes

Proposal list (student/postdoc-led indicated with [†]):

Millimeter: 39 proposals, 10 programs. Telescopes: SMA (425 hr), NOEMA (239.4 hr), ALMA (50.1 hr).

- *AT2018cow: the poster-child relativistic explosion for high-frequency time-domain astronomy*
SMA, $\approx$ 76 hr, DDT, 2018A
ALMA, 2.6 hr, DDT, 2017
- *Single-object DDTs: rapidly rising blue transients discovered by ZTF*
SMA, $\approx$ 5 hr, DDT, 2018A
NOEMA, 8 hr DDT D20, 4 hr DDT D20, 1 hr DDT D24
ALMA, 5.6 hr DDT 2022
- *Landscape of relativistic stellar explosions (LLGRBs, dirty fireballs)*
SMA, 96 hr 2018B, 18 hr 2021B, 6 hr 2022A, 18 hr 2022B
NOEMA, 22.1 hr W21, 19.9 hr W22, 20.8 hr S23, 10.5 hr W23
- *The death throes of massive stars, revealed through early millimeter observations*
SMA, 36 hr 2019A, 18 hr 2019B
NOEMA, 32 hr S19, 14 hr W24EX, 9.6 hr S25CX, 8.8 hr W25EK
ALMA, 9.7 hr 2019
- *A new class of energetic stellar explosions in a dense medium*
SMA, 36 hr 2020A, 30 hr 2020B, 6 hr 2021A
NOEMA, 6 hr W20BO, 9 hr S21AY, 18.8 hr W24EW, 10.6 hr S25CW
ALMA, 6.9 hr 2023, 6.9 hr 2024, 13.8 hr 2025 all through joint VLA
- *DDT: Continued Monitoring of Relativistic Jetted TDE AT2022cmc*
SMA, 54 hr DDT, 2021B
- *Millimeter-wavelength Monitoring of a Galactic Nova*
SMA, 6 hr 2024B, 6 hr 2025A, 14 hr 2025B
- *GRB 250419A: A Candidate Dirty Fireball*
NOEMA, 2.8 hr DDT, E24AN
- *The first millimeter observations of an orphan afterglow*
ALMA, 2.3 hr DDT, 2022
- *A Candidate Fast Extragalactic Transient Discovered at Millimeter Wavelengths*
ALMA, 2.3 hr DDT, 2022

Radio: 40 proposals, 16 programs. Telescopes: VLA (388.3 hr), ATCA (256 hr), VLBA (27 hr), EVN (23 hr), MeerKAT (16.5 hr), uGMRT (3 hr).

- *Monitoring the flux from eclipsing binary pulsar Terzan 5A*
VLA, 4 hr, 2013A
- *Short soft transients from MAXI*
VLA, 2 DDTs, each 3 hr, 2017A & 2017B
- *Relativistic explosions in dense circumburst matter (luminous fast blue optical transients)*
VLA, 206.4 hr total spanning 2019A–2026B
ATCA, 216 hr total spanning 2021–2022
- *Search for a radio counterpart to eROSITA X-ray transients*
VLA, 2 hr, 2019B, DDT

- *A search for off-axis jets in Ic-BL supernovae with GRB-like host environments*
VLA, 16 hr, 2020A
- *VLA+Chandra observations of Ic-BL SNe with ZTF high-cadence light curves*
VLA, 16 hr, 2021B
- *Expanding the known range of LFBOT physical properties (single-object DDTs)*
VLA, 37.1 hr, DDTs spanning 2023A–2025A
uGMRT, 3 hr, DDT, 2019
- *VLA characterization of Einstein Probe fast X-ray transients*
VLA, 2 DDTs in 2024A, each 2 hr
- [†] *Confirming an Off-Axis Jet in PTF10tqv, a Ic-BL SN*
VLA, 1.25 hr, 2024B, DDT; then 1.17 hr, 2025B
MeerKAT, 1 hr, DDT, 2025
EVN, 11 hr, 2025
- *Supernovae with Luminous Radio Counterparts: A Systematic ZTF+VLASS Search*
VLA, 20.61 hr, 2025A
- *SN2025mkn: A Likely Lensed Fast Transient at High Redshift*
VLA, 2.5 hr, 2025B
- [†] *Searching for Off-axis Jets in SNe Ic-bl (large-scale ZTF search)*
VLA, 71.38 hr, 2026A
- *X-ray Flashes: Dirty Fireballs or Off-axis GRBs?*
VLA (joint with XMM-Newton and HST), 17.03 hr, 2026A
- [†] *Late Radio Brightening in the LFBOT AT2024aehp: An Off-axis Jet?*
EVN, 12 hr, 2025
- *Constraining the Origin of Extragalactic Transients Discovered by Wide-field Millimeter Surveys*
ATCA, 40 hr, 2024OCTS
MeerKAT, 4 hr, DDT, 2025
[†] MeerKAT, 11.3 hr, 2025
- *Confirmation of Superluminal Motion for Rare Relativistic Supernova AT2018cow*
VLBA, 12 hr, 2018A
- *Search for an off-axis jet in a broad-lined Ic supernova*
SN2020bvc: VLBA, 12 hr, 2020A
[†]SN2025shf: VLBA, 3 hr, 2026A

X-ray: 5 proposals, 5 programs. Chandra (100 ksec=27.8 hr), Swift (42 ksec=11.7 hr), XMM-Newton (40 ksec=11.1 hr, VLA as lead observatory). Also 56 *Swift* ToO requests totaling ≈ 220 ksec=62 hr.

- *A Candidate Low-luminosity Gamma-ray Burst Identified by an Early Optical Flash*
Chandra, 20 ksec, DDT, Cycle 21
- *VLA+Chandra Observations of Ic-BL Supernovae with ZTF High-Cadence Light Curves*
Chandra, 20 ksec, DDT, Cycle 22
- *Discovery of Minute-timescale Flares in the Aftermath of a Cosmic Explosion*
Chandra, 40 ksec, DDT, Cycle 23
- *The X-ray Flash 250419A: A Candidate Dirty Fireball*
Chandra, 20 ksec, DDT, Cycle 26
- *The Landscape of Relativistic Stellar Explosions*
Swift, 42 ksec, Cycle 19

UVOIR: 17 proposals, 7 programs. Telescopes: JWST (2 orbits = 16 hours), HST (6 orbits + additional 5 orbits with VLA as lead observatory = 17.6 hr), Gemini North (15 hr), Gemini South (29.3 hr), Palomar 60-inch (16.5 hr), Palomar 48-inch (3 nights)

- *AT2018cow: A Unique Laboratory for Super-Eddington Accretion and Black-hole Growth*
JWST, 2 orbits (16 hr), Cycle 5
- *Tracking the Emergence of an Accretion Disk in an LFBOT*
HST, 6 orbits (9.6 hr), DDT, Cycle 32

- *A Rapid Response to the Youngest ZTF Explosions*
Gemini South, 4.4 hr, 2021A
- *Finding Relativistic Stellar Explosions as Fast Optical Transients*
Gemini North, 15 hr from 2022A–2026A
Gemini South, 21.8 hr from 2022A–2026A
- *Characterizing a Fast Extragalactic Transient Discovered by a Millimeter Survey*
Gemini South, 3.14 hr, Fast Turnaround, 2025A
- *The First Optically Selected Population of Relativistic Afterglows*
Palomar 60-inch/SEDM, 8.25 hr each of 2019A and 2020A
- *ZTF Experiment Proposal: Monitoring an ULTRASAT High-cadence Field*
Palomar 48-inch/ZTF, 3 nights (≈ 24 hr), 2024

LEADERSHIP & PROFESSIONAL SERVICE

Scientific leadership and membership roles; WG=Working Group

2025–Present	ALMA2040 transients & time-domain astronomy WG (Member)
2024–Present	Africa Millimetre Telescope (Science Team Member)
2024–Present	South Pole Telescope (Full Senior Member)
2023–Present	Ultraviolet Transient Astronomy Satellite (ULTRASAT; Participating Scientist)
	• (2023–Present) Co-chair, Gamma-ray Bursts (GRB) WG
2023–Present	Atacama Cosmology Telescope (Senior Member)
2023–Present	Simons Observatory (Senior Member)
	• (2026–Present) Member, committee on ToO interrupts
2022–Present	CCAT Observatory
	• (2022–Present) Co-lead, Time-domain Astrophysics WG
2021–Present	UltraViolet EXplorer (UVEX) Mission (Co-I). <i>NASA MIDEX: launch 2030</i>
2018–Present	Zwicky Transient Facility (ZTF; Member)
	• (2025–Present) Co-chair, Supernova & Relativistic Explosions WG
	• (2023–Present) Member, ZTF Science Steering Committee
2022–2025	CMB-S4 (Senior Member)
	• (2021–2023) Co-chair, Sources & Transients WG
2020–2023	DESI (member 2020–2022; continuing collaborator 2022–2023)

Invited positions for service external to Cornell

2026	Ambassador, Aspen Center workshop on “Physics and Astrophysics of Neutrino-Dense Environments”
2025	Subaru Time Allocation Committee
2025	NSF Review Panel
2025	SOC, NASA’s 4th TDAMM Workshop
2025	Internal Reviewer, Simons Observatory time-resolved pipeline
2025	External Examiner, Viva at University College Dublin
2023–Present	Member, ZTF Publication Board
2023–Present	Member, ZTF Science Steering Committee
2022–Present	Member, Science Advisory Council, DSA-2000
2019–Present	Referee/reviewer for ApJ, ApJL, MNRAS, Nature Astronomy, <i>Science</i> , A&A, PASP
2023–2024	Member, SOC, Workshop on Fast Extragalactic Transients, Bormio, Italy
2023	Review Panelist, NASA ROSES Program
2023	Reviewer, James Webb Space Telescope DD proposal
2023	Reviewer, Hubble Space Telescope DD proposal
2023	Member, SOC, CMB-S4 Collaboration Meeting
2022–2023	Member, Science Organizing Committee, “Scientific Frontiers and Synergies for the

	DSA-2000 Radio Camera” Conference
2022	Member, Miller Annual Symposium Organizing Committee
2022	Member, SOC, CMB-S4 Collaboration Meeting
2022	Member, SOC, Workshop on “Astrophysics with the CMB-S4 Survey”

Other external service

2021–2022	Co-organizer, Theoretical Astrophysics Center Seminar Series, UC Berkeley
2021–2022	Organizer, Explosive Astronomy Seminar Series, UC Berkeley
2021	Member, Chandra Time Allocation Committee
2021	Member, Gemini Time Allocation Committee
2021	Co-organizer, Session on Transients with CMB-S4, CMB-S4 Summer Workshop
2021	Reviewer, NASA FINESST Program
2021	Time Allocation Committee: Gemini, Chandra
2021	Peer Reviewer, ALMA proposals
2021	Co-organizer, Cal-URSA Research Program
2021	Organizer, Workshop on Status of Millimeter-Transient Searches (virtual)

TEACHING

At Cornell

ASTRO 2211: Stars, Galaxies, and Cosmology (undergraduate course)

- Fall 2026: 45 students.
- Fall 2025: 40 students. Course rating: 4.68/5. Instructor rating: 4.68/5
- Fall 2024: 34 students. Course rating: 4.52/5. Instructor rating: 4.72/5
- Fall 2023: 41 students. Course rating: 4.79/5. Instructor rating: 4.87/5

ASTRO 4432/6530: Astrophysical Processes (joint undergraduate+graduate course)

- Spring 2026. 12 students. Course rating: 4.75/5. Instructor rating: 4.75/5
- Spring 2025: 17 students. Course rating: 5.00/5. Instructor rating: 4.92/5
- Spring 2024: 10 students. Course rating: 5.00/5. Instructor rating: 4.90/5

ASTRO 7683: Seminal Papers in Astronomy and Planetary Science (graduate course)

- Spring 2023: 10 students. Course rating: 5.00/5. Instructor rating: 5.00/5

ASTRO 4940, PHYS 2298, PHYS 4498/4499: Independent Study or Senior Thesis (undergraduates)

- 2 students each semester from Fall 2022–Spring 2024

Workshops

- 2025: Instructor, Center for Computational Astrophysics LSST Summer School
- 2018: Instructor, ZTF Summer School
- 2016: Lead Instructor, Gemini Observatory Workshop on Data-Driven Modeling of Spectra

ADVISING AND MENTORING

Postdoctoral Scholar Mentoring

Fall 2026–	Laura Cotter: X-ray flashes
Fall 2025–	K-Ryan Hinds (joint with Caltech): fast transients with ZTF & LSST
Fall 2025–	Rahul Jayaraman, Cornell Klarman Fellow: fast transients with TESS
2024–2026	Genevieve Schroeder (Cornell Postdoctoral Scholar → SKA Scientist)

Graduate Students

Fall 2026–	Nitya Nigam (1st year): collapsar landscape, connecting theory & observations
Fall 2024–	Michael Camilo (3rd year): radio modeling of trans-relativistic explosions

Fall 2023– Cassie Sevilla (4th year): fast blue optical transients
 2025–2026 Gokul Srinivasaragavan (UMD Grad → CIERA Postdoctoral Fellow, Northwestern)

Undergraduate Students

2026– Maya Fanti (Ithaca College Physics): ZTF×DESI
 2025– Beilin Liu (Cornell Computer Science): A web app for transient classification
 2025– Amaliya Atamalibekova (Cornell Physics): A search for sub-threshold γ -ray bursts
 2024– Marquice Sanchez-Fleming (Cornell junior, physics): SN mm-wave counterparts
 2024–2026 Jack Pope (Cornell senior, physics & data science): SN cm-wave counterparts
 2023–2025 Jada Vail (Cornell physics → Epic Systems): McNair Scholar, Nexus Scholar
 2024–2025 Harlan Phillips (REU student): A catalog of all ZTF extragalactic transients
 2022–2024 Maggie Li (Cornell physics → PhD student at Caltech)
 • 2023 Roger & Mary Lou West Fellow; 2024 AAS Chambliss Award; 2024 Shelley Undergraduate Research Award in Astronomy; 2024 Bethe Thesis Prize in Physics
 Summer 2023 Joshua Grajales (REU student, Columbia senior): ML identification of FBOTs
 2022–2023 Kailai Wang (Cornell physics): photometry of early GRB optical afterglows
 2022–2023 William Hohensee (UC Berkeley astrophysics): DESI observations of ZTF host galaxies
 Summer 2022 Mary Gerhart (UC Berkeley astrophysics): FBOT identification in survey data
 Summer 2021 Alexis Andersen, Autumn Awbrey, Ruby Wong (UC Berkeley physics/astrophysics): DESI observations of transient host galaxies (co-advised with Peter Nugent)
 Summer 2021 Caitlin King (Northern Arizona University): orphan afterglow model predictions
 DOE Science Undergraduate Laboratory Internship (SULI) program

ACADEMIC PRESENTATIONS

Summary: 91 total (67 invited), of which 39 at international conferences (24 invited), 49 department colloquia/seminars (39 invited), 2 summer schools (1 as lead organizer), 2 panels (1 press)

Selected recent invited talks:

2027 Colloquium, University of California Los Angeles, Los Angeles CA, USA
 2027 Colloquium, Black Hole Initiative (BHI), Harvard University, Cambridge MA, USA
 2027 Newton Lacy Pierce Prize Lecture, AAS Annual Winter Meeting, Salt Lake City UT, USA
 2026 AstroLunch, University of Pittsburgh & CMU, Pittsburgh PA, USA
 2026 Colloquium, University of Illinois Urbana Champaign, Urbana Champaign IL, USA
 2026 Plenary, Conference on “The Dynamic Radio Sky,” McGill University, Montreal, Canada
 2026 Plenary, Conference on “Explosive Astrophysics in the Era of High-Cadence Astronomy,” Purdue University, Lafayette IN, USA
 2026 Plenary, IAU Symposium on “Future landscape of astrophysical transients: novel approaches in theory and observations,” Turku, Finland
 2026 Plenary, Conference on “RAPID Response: Hot-wiring the Next Generation of Time-Domain Science,” Caltech, Pasadena CA, USA
 2026 Plenary, “Multi-Messenger Astrophysics in the Dynamic Universe” Workshop, Kyoto, Japan
 2025 Astronomy & Astrophysics Seminar, New York University, New York USA
 2025 GRB Panel & SN Overview, NASA’s 4th TDAMM Workshop, Huntsville AL, USA
 2025 Colloquium, Max Planck Institute for Radio Astronomy, Bonn Germany
 2025 Plenary, Science Meeting for the Black Hole Explorer (virtual)
 2025 Plenary, “Shaping the Future of Time-Domain Astronomy with LSST” Conference, Rio de Janeiro, Brazil
 2025 Talk, “PRIMA and the Future of Far-IR Science” Workshop, Caltech, Pasadena CA, USA
 2025 Colloquium, Goddard Space Flight Center, Greenbelt MD, USA
 2025 Trotter Space Institute Astronomy Seminar, McGill University, Montreal QC, Canada
 2025 Colloquium, Syracuse University, Syracuse NY, USA
 2025 Seminar, University College Dublin, Dublin, Ireland

2025 Colloquium, Herzberg Astronomy and Astrophysics Research Centre, Victoria BC, Canada
 2025 Colloquium, University of British Columbia, Vancouver BC, Canada
 2024 Plenary, “MAXI 15 Year Workshop for the Time Domain Astronomy,” Tokyo, Japan
 2024 Colloquium, Pennsylvania State University, College Park PA, USA
 2024 Colloquium, Aspen Center for Physics, Aspen CO, USA ([recording](#))
 2024 Plenary, Current Themes Workshop, Niels Bohr Institute, Copenhagen, Denmark
 2024 Plenary, “Cosmic Transients in the Era of Large Surveys” Symposium, Swedish Royal Academy of Sciences, Stockholm, Sweden
 2024 Plenary on Transient Science, CMB-S4 Spring Collaboration Meeting (virtual)
 2024 Colloquium, Princeton/IAS, Princeton NJ, USA
 2023 Plenary, 32nd Texas Symposium on Relativistic Astrophysics, Shanghai, China
 2023 Colloquium, MIT Astrophysics, Cambridge MA, USA
 2023 Colloquium, UVA/NRAO, Charlottesville VA, USA
 2023 Plenary, The Transient and Variable Universe Conference, UIUC, Illinois, USA
 2023 Talk, CMB-S4 Spring Collaboration Meeting (virtual)
 2023 Colloquium, Columbia University, New York NY, USA
 2023 Review, MIAPbP “Interacting Supernovae” Workshop, Garching, Germany
 2023 Colloquium, University of Toronto, Toronto ON, Canada
 2022 Colloquium, Caltech, Pasadena CA, USA
 2022 Talk, Time Domain and Multi-Messenger Astrophysics NASA Workshop (virtual)
 2022 Plenary, CMB-S4 Collaboration Meeting (virtual)
 2022 Talk, AAS Special Session on “An Update on Astrophysics and Cosmology from Cosmic Microwave Background Measurements in the Next Decade”
 2022 Colloquium, Radboud University, Nijmegen, Netherlands
 2022 Colloquium, Carnegie Observatories, Pasadena CA, USA
 2022 Talk, APS April Meeting, Cecilia Payne-Gaposchkin Dissertation Award Finalist
 2022 CCAT-Prime/FYST Collaboration Meeting (virtual)
 2022 Tor Vergata Astrophysics Seminar (virtual)
 2022 Special Physics & Astronomy Seminar, Northwestern University, Evanston IL, USA
 2022 Colloquium, U.T. Austin, Austin TX, USA
 2022 Colloquium, Cornell University, Ithaca NY, USA

SELECTED COMMUNITY ENGAGEMENT

2026 Speaker, Cornell Astronomical Society
 2026 Lecture on the P200, Corning Museum of Glass, Corning, NY, USA ([recording](#))
 2025 Lecturer, Warrior Scholars Program, Cornell
 2025 Volunteer, Spacecraft Planetary Image Facility Open House, Cornell
 2025 Guest Speaker, Ithaca High School
 2025 Speaker, Ithaca Astro on Tap
 2025 Interviewed for Diaries of the Cosmos, part of NASA’s Universe of Learning
 2025 Lecture for Astronomy Section, Rochester Academy of Science
 2024 Interviewed for class project, Irvine Valley College
 2024 Lecture for Cornell Friends of Astronomy
 2024 Lecture for North Jersey Astronomical Society ([recording](#))
 2023 Speaker, Cornell Astronomical Society
 2023 Interviewed for Hotel Mars CBS Eye On The World podcast
 2023 Interviewed for Science Review magazine, American School in London
 2023 Member, Career Panel, 4-H Camp at Cornell
 2023 Interviewed for graduate student podcast *Cosmos Crusaders* ([recording](#))
 2022 Keynote Speaker, Annual Cray User Group Meeting, Monterey CA
 2017–2022 Interviewer, MIT Admissions
 2021 Compass Lecture, UC Berkeley

2021	Speaker, Riverside Astronomical Society (virtual)
2019	Speaker, Greenway Talk Series, Palomar Observatory
2019	Speaker, Owens Valley Radio Observatory Lecture Series
2019	Speaker, Caltech Graduate Research Spotlight
2019	Speaker, Ventura County Astronomical Society
2019	Speaker, Greenway Talk Series, Palomar Observatory
2018	Contributing Writer, Caltech Letters
2018	Speaker, College of the Canyons Star Party
2018	Visitor, 8th-grade class, St. Philip the Apostle School, Pasadena CA
2017	Volunteer, Orbit Deep Learning Days, Huntington Library, Pasadena CA
2017	Speaker, Astro on Tap, Pasadena CA
2017	Speaker, Riverside Astronomical Society
2017	Speaker, Ventura County Astronomical Society
2017	Speaker, High School Summer Camp, Culver City CA
2016	Speaker, Santa Monica Astronomy Club
2016	Volunteer, Field Trip, iChicas After-school Program
2015	Speaker, St. Philip Reverse Science Fair, Pasadena CA
2015	Volunteer, Webster Elementary Science and Stargazing Night, Pasadena CA